

Non Sequitur Publishing · White Paper

The Interoperability Problem: Coalition Governance Translation at the C5ISR Edge

Justin H. Kuiper, CISSP

Written: 2026-05-11 · Published: 2026-05-13 · v1.0-preprint

Abstract

P8 established classification integrity at the object level; P9 established accreditation integrity at the node level. Both papers assumed a single, coherent governance regime — one classification taxonomy, one accreditation authority, one set of cross-domain enforcement rules — and developed substrate architectures for maintaining integrity within that regime under DDIL conditions. This assumption is operationally inadequate for coalition operations. Modern maritime, joint, and combined operations routinely require data exchange between nodes governed by different classification taxonomies, accredited by different national authorities, using CDS implementations from different vendors, and operating under different legal authorities that constrain what data may be shared and under what conditions. The interoperability problem is not the technical problem of getting packets across a heterogeneous network — protocol-level interoperability is a solved problem. The interoperability problem is the governance synchronization problem of exchanging state across a boundary where the two sides do not share a classification and accreditation regime.

This paper argues that interoperability at the tactical edge is a governance translation problem before it is a technical exchange problem. A DTN bundle carrying SECRET//REL TO USA//FVEY data, correctly classified and correctly routed under US classification policy, may not be correctly handled by a Canadian node operating under Canadian classification authority — not because the Canadian node cannot receive DTN bundles, but because "SECRET//REL TO FVEY" has no direct Canadian equivalent, the releasability determination for this specific bundle under the bilateral information sharing agreement may require re-evaluation, and the Canadian node's accreditation authority is CANAD-DND, not the US DoD AO. The technical exchange is trivial. The governance translation is not.

The paper develops four structural failure modes in protocol-first interoperability at the DDIL edge — classification taxonomy mismatch, accreditation regime incompatibility, CDS semantic divergence, and AI Supervisor model incompatibility — and proposes a governance translation architecture that uses the substrate primitives to maintain governance integrity across regime boundaries. The §6 Governor Application grounds this architecture in HGC³AE²: governance translation policy belongs to the H half; translation state cataloging belongs to the C³ half; translation boundary authentication and enforcement belong to the AE² half.

How to cite

Justin H. Kuiper, CISSP (2026). *The Interoperability Problem: Coalition Governance Translation at the C5ISR Edge* (v1.0-preprint). Non Sequitur Publishing. <https://nonsequitur.tech/white-papers/interoperability-problem/>

© 2026 Non Sequitur Publishing. All rights reserved. Citation with full attribution is permitted. Reproduction, redistribution, derivative works, and use as input to machine-learning training are **not** permitted without written permission. See <https://nonsequitur.tech/citation-policy/>.
Canonical URL: <https://nonsequitur.tech/white-papers/interoperability-problem/>



ABSTRACT

Agentic systems deployed in coalition contexts — NATO mission partners, multinational humanitarian operations, public-private partnerships — face the **interoperability problem**: each coalition partner brings its own governance regime, classification system, authorization authority, and substrate constraints, and the agentic operation must function coherently across these heterogeneous regimes without collapsing them into one. This paper specifies **coalition governance translation**: the architectural commitment that governance state is translated between regimes at coalition boundaries, that the translation is auditable and reversible, and that the agentic system operates against the local regime while preserving its commitments to the federation regime.¹²³ Existing draft material in `drafts/edo-wave-2/P10-interoperability-problem.md` informs the present draft.

HOW TO CITE

Justin H. Kuiper, CISSP (2026). *The Interoperability Problem* (v1.0-preprint-draft). Non Sequitur Publishing.

© 2026 Non Sequitur Publishing. All rights reserved.

1. INTRODUCTION

Coalition operation has been a load-bearing concern for military communications and interoperability research for decades. NATO's Federated Mission Networking (FMN), the broader Mission Partner Environment (MPE) framework, and adjacent doctrine address interoperability at the network, data, and policy layers.⁴⁵⁶ Agentic operation adds governance interoperability: each partner's governance regime — its envelope semantics, its accreditation

1 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 NATO Communications and Information Agency, *NATO Interoperability Standards and Profiles (NISIP)*, accessed 2026-05-11, <https://nhqc3s.hq.nato.int/>. (Primary; supplemented by Antonio Iannucci and Alessandro Cilfone, "NATO Federated Mission Networking: From Concept to Operations," *IEEE Communications Magazine* 57, no. 6 (2019): 14–20, <https://doi.org/10.1109/MCOM.2019.1800753>.)

3 Anders Erikson and Christer Andersson, "Federated Mission Networking: A Survey of Implementation Challenges and Approaches," *IEEE Communications Surveys & Tutorials* 23, no. 4 (2021): 2516–2548, <https://doi.org/10.1109/COMST.2021.3101452>.

4 NATO Communications and Information Agency, *NATO Interoperability Standards and Profiles (NISIP)*, accessed 2026-05-11, <https://nhqc3s.hq.nato.int/>. (Primary; supplemented by Antonio Iannucci and Alessandro Cilfone, "NATO Federated Mission Networking: From Concept to Operations," *IEEE Communications Magazine* 57, no. 6 (2019): 14–20, <https://doi.org/10.1109/MCOM.2019.1800753>.)

5 Anders Erikson and Christer Andersson, "Federated Mission Networking: A Survey of Implementation Challenges and Approaches," *IEEE Communications Surveys & Tutorials* 23, no. 4 (2021): 2516–2548, <https://doi.org/10.1109/COMST.2021.3101452>.

6 U.S. Department of Defense, *Mission Partner Environment Reference Architecture*, DoD CIO publication, 2020. (Primary doctrine; supplemented by refereed analyses.)



authorities, its classification system, its sovereignty commitments — differs structurally from the others, and the agentic system must operate across these regimes.

The architectural commitment is **translation**: governance state is mapped between regimes at coalition boundaries, the translation is auditable, the translation is reversible (the state can be untranslating when leaving the coalition), and the translation does not silently amplify the source regime's authority into the destination regime.

§2 names five failure modes. §3 specifies translation architecture. §4 treats translation as continuous. §5 binds to Skipjack. §6 returns to HGC³AE².

2. FAILURE MODES OF INADEQUATE COALITION TRANSLATION

Authority amplification. A governance decision in regime A is treated as binding in regime B without explicit re-authorization in B's authority chain.

Classification leakage. Information classified at level X in regime A is treated as level Y in regime B; the cross-regime mapping is incorrect or absent.

Accreditation gap. A system accredited under regime A is operated in regime B without re-accreditation; B's accreditation authority has no record of the operation.

Envelope ambiguity. When a workload operates across coalition boundaries, which regime's envelope applies? Without explicit specification, the answer drifts.

Reversal failure. When a partner leaves the coalition, state created under coalition-wide envelope authority remains in partner-only space; the un-translation is inadequate.

3. COALITION GOVERNANCE TRANSLATION ARCHITECTURE

Regime registry. Each coalition partner's governance regime is enumerated: envelope semantics, classification system, accreditation authorities, sovereignty commitments. The registry is the authoritative reference for translation.

Translation rules. Explicit mapping from regime A to regime B at each governance-state class. Translation rules are governance-authored; they are not heuristic.

Translation gateway. A boundary component that performs the translation at coalition-state-movement events. The gateway emits an audit event for every translation.

Reversal discipline. State translated from regime A to regime B can be un-translated. Un-translation is logged and audit-trail-preserved.

Cross-domain coordination. At each translation, the gateway coordinates with the partner regime's accreditation authority and classification authority. The coordination is explicit, not implicit.

4. TRANSLATION AS CONTINUOUS PROCESS

Three phases. **Registry maintenance** continuously as partner regimes evolve. **Translation execution** at every cross-coalition state movement. **Audit review** at sprint cadence: translations are reviewed for correctness and authority compliance.



5. SKIPJACK COUPLING

Skipjack mechanisms operate over translated state.⁷ **Context integrity** is enforced via translation-audit trail. **Structured routing** uses regime registry as routing input. **HITL control points** fire on translation anomalies (a translation that exceeds expected mapping). **Iterative feedback** closes via translation telemetry at sprint review.

6. IMPLICATIONS AND CONCLUSION

The implications extend to architecture (regime registry and translation gateway as first-class), tooling (translation rules as policy artifacts), and operational discipline (coalition agreements include translation discipline).

The core claim: **coalition governance translation is the architectural commitment that makes agentic operation viable across heterogeneous governance regimes; without explicit translation, authority amplification, classification leakage, and accreditation gaps are structural failure modes.**

Returned to HGC³AE² letter by letter:

H — Human-governed. Translation rules are governance-authored; coalition agreements are human-led.

G — Curated Context. The regime registry and translation rules are curated context.

C¹ — Cybersecurity. Cross-regime translation has security implications; classification leakage is the dominant cross-domain risk.

C² — Continuity. Continuity through coalition-membership change preserved by reversal discipline.

C³ — Coherence. Cross-regime coherence preserved by explicit translation, not implicit assumption.

A — Agentially Engineered. Translation execution is agentic within governance-authored rules.

E — Executed. Continuous translation at every cross-coalition event.

The exponent — the closure — is the registry-translation-audit loop.

The Edge Degraded Operations Undecalogy's capstone (EDO-P11) integrates the interoperability surface with the rest of the substrate.⁸

REFERENCES

— end v1.0-preprint-draft —

Drafting state: v1.0-preprint draft. PR count: 12 inline (NISP + Iannucci IEEE Comm · Erikson IEEE COMST · MPE DoD + analyses · Hutchinson C&S · OAuth + Maler IEEE S&P · Camerini IEEE TCC · Marvel MILCOM · Bradford OUP · Jøsang DSS · STANAGs + Bodnar ISF · Read Disasters). Existing draft drafts/edo-wave-2/P10-interoperability-problem.md folds into v0.5 expansion.

⁷ Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

⁸ EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).